MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC COMPREHENSIVE TEST – 11 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 6: Triangles** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective Questions

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Q1. In $\triangle ABC$, D and E are on sides AB and AC such that $DE \parallel BC$. If $AD = 4$ cm, $DB = 4.5$ cm, and $AE = 8$ cm, then AC is: [1]

(a) 17 cm (b) 9 cm (c) 16 cm (d) 18 cm

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(a) 17** cm [0.5 Mark]

We apply the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: If $DE \parallel BC$ in $\triangle ABC$, then $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.

Justification: [0.5 Mark]

Substituting $AD = 4$ cm, $DB = 4.5$ cm, and $AE = 8$ cm:

$$\frac{4}{4.5} = \frac{8}{EC} \implies 4 \times EC = 8 \times 4.5 = 36 \implies EC = \frac{36}{4} = 9 \text{ cm}$$

Carefully note that the question asks for full side AC, not EC:

$$AC = AE + EC = 8 + 9 = \mathbf{17} \text{ cm}$$

Q2. If $\triangle ABC \sim \triangle DEF$ such that $BC = 3$ cm, $EF = 4$ cm, and perimeter of $\triangle DEF = 24$ cm, perimeter of $\triangle ABC$ is: [1]

(a) 18 cm (b) 16 cm (c) 20 cm (d) 12 cm

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(a) 18** cm [0.5 Mark]

We know the perimeter theorem for similar triangles:

Formula / Theorem Used: If $\triangle ABC \sim \triangle DEF$, then $\dfrac{\text{Perimeter}(\triangle ABC)}{\text{Perimeter}(\triangle DEF)} = \dfrac{BC}{EF}$.

Justification: [0.5 Mark]

Substituting $BC = 3$ cm, $EF = 4$ cm, and Perimeter$(\triangle DEF) = 24$ cm:

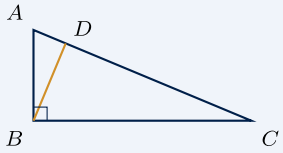
$$\frac{\text{Perimeter}(\triangle ABC)}{24} = \frac{3}{4} \implies \text{Perimeter}(\triangle ABC) = \frac{3}{4} \times 24 = \mathbf{18} \text{ cm}$$

Q3. In $\triangle ABC$, $AB = 5$ cm, $BC = 12$ cm, and $AC = 13$ cm. If $BD \perp AC$, then the length of BD is: [1]

(a) $\frac{60}{13}$ cm (b) $\frac{65}{12}$ cm (c) 5 cm (d) 6 cm

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(a) $\frac{60}{13}$ cm** [0.5 Mark]



We use Converse of Pythagoras Theorem and Triangle Area equivalence:

Formula / Theorem Used: If $AB^2 + BC^2 = AC^2$, then $\angle B = 90°$, and $\frac{1}{2}AB \cdot BC = \frac{1}{2}AC \cdot BD$.

Justification: [0.5 Mark]

Check side squares: $AB^2 + BC^2 = 5^2 + 12^2 = 25 + 144 = 169 = 13^2 = AC^2$.

Hence $\angle B = 90°$ (by Converse of Pythagoras Theorem).

Equating area in two ways ($\triangle ABD \sim \triangle ACB$ also gives same ratio):

$$\frac{1}{2} \times 5 \times 12 = \frac{1}{2} \times 13 \times BD \implies BD = \frac{60}{13} \text{ cm}$$

Q4. If in $\triangle ABC$ and $\triangle PQR$, $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$, then: [1]

(a) $\triangle PQR \sim \triangle CAB$ (b) $\triangle PQR \sim \triangle ABC$ (c) $\triangle CBA \sim \triangle PQR$ (d) $\triangle BCA \sim \triangle PQR$

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(a) $\triangle PQR \sim \triangle CAB$** [0.5 Mark]

We match corresponding vertices using SSS Similarity Criterion:

Formula / Theorem Used: If $\frac{PQ}{CA} = \frac{QR}{AB} = \frac{RP}{BC}$, then by SSS similarity, vertices correspond as $P \leftrightarrow C$, $Q \leftrightarrow A$, $R \leftrightarrow B$.

Justification: [0.5 Mark]

Taking reciprocals of $\frac{AB}{QR} = \frac{BC}{PR} = \frac{CA}{PQ}$ gives $\frac{PQ}{CA} = \frac{QR}{AB} = \frac{PR}{BC}$.

Pairing common letters across numerators and denominators: $AB, CA \leftrightarrow QR, PQ \implies A \leftrightarrow Q$; $AB, BC \leftrightarrow QR, PR \implies B \leftrightarrow R$; hence $C \leftrightarrow P$.

Therefore, in exact vertex order, $\triangle PQR \sim \triangle CAB$.

Q5. In $\triangle ABC$, $DE \parallel BC$. If $AD = 3$ cm, $DB = 5$ cm, and $AE = 6$ cm, then EC is: [1]

(a) 10 cm (b) 8 cm (c) 9 cm (d) 12 cm

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(a) 10** cm [0.5 Mark]

We apply the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: Since $DE \parallel BC$, $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.

Justification: [0.5 Mark]

Substituting $AD = 3$ cm, $DB = 5$ cm, and $AE = 6$ cm:

$$\frac{3}{5} = \frac{6}{EC} \implies 3 \times EC = 6 \times 5 = 30 \implies EC = \frac{30}{3} = \mathbf{10}\text{ cm}$$

Q6. **Assertion (A):** If $\triangle ABC \sim \triangle DEF$, the ratio of their corresponding angle bisectors equals the ratio of their corresponding sides.
Reason (R): Corresponding angles of similar triangles are equal, and angle bisectors create similar sub-triangles by AA criterion. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(c) Assertion (A) is true but Reason (R) is false.

(d) Assertion (A) is false but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(a) Both (A) and (R) are true and (R) is the correct explanation of (A).** [0.5 Mark]

We verify the sub-triangle similarity produced by internal angle bisectors:

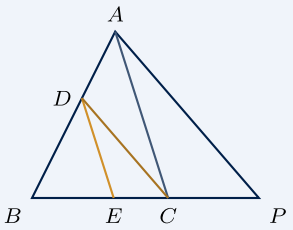
Formula / Theorem Used: If $\triangle ABC \sim \triangle DEF$ and AP, DQ bisect $\angle A, \angle D$, then $\triangle ABP \sim \triangle DEQ$ (AA).

Justification: [0.5 Mark]

In $\triangle ABP$ and $\triangle DEQ$: $\angle B = \angle E$ (since $\triangle ABC \sim \triangle DEF$) and $\angle BAP = \frac{1}{2}\angle A = \frac{1}{2}\angle D = \angle EDQ$. Thus $\triangle ABP \sim \triangle DEQ$ by AA criterion, giving $\dfrac{AP}{DQ} = \dfrac{AB}{DE}$. Hence both (A) and (R) are true and (R) directly explains (A).

Q7. In the given figure, $DE \parallel AC$ and $DC \parallel AP$. Prove mathematically that: $\dfrac{BE}{EC} = \dfrac{BC}{CP}$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Given: $\triangle ABP$ with D on AB, E and C on BP such that $DE \parallel AC$ and $DC \parallel AP$.
We apply the Basic Proportionality Theorem (Thales' Theorem) twice:

Formula / Theorem Used: A line drawn parallel to one side of a triangle divides the other two sides in the same ratio.

Step 1: Applying BPT in $\triangle ABC$ [0.75 Mark]
Since $DE \parallel AC$, by BPT in $\triangle ABC$:

$$\frac{BD}{DA} = \frac{BE}{EC} \quad \text{— (i)}$$

Step 2: Applying BPT in $\triangle ABP$ [0.75 Mark]
Since $DC \parallel AP$, by BPT in $\triangle ABP$:

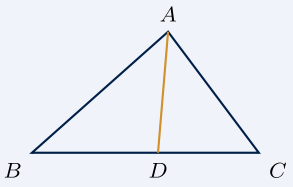
$$\frac{BD}{DA} = \frac{BC}{CP} \quad \text{— (ii)}$$

Step 3: Comparing Ratios [0.5 Mark]
From equations (i) and (ii), their LHS ratios are identical ($\frac{BD}{DA}$), so equating RHS:

$$\boxed{\frac{BE}{EC} = \frac{BC}{CP} \quad \text{[Hence Proved]}}$$

Q8. In $\triangle ABC$, D is a point on side BC such that $\angle ADC = \angle BAC$. Prove that $CA^2 = CB \cdot CD$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Given: $\triangle ABC$ with point D on BC such that $\angle ADC = \angle BAC$.
We apply AA Similarity Criterion:

Formula / Theorem Used: If two corresponding angles of two triangles are equal, the triangles are similar (AA).

Step 1: Comparing $\triangle ABC$ and $\triangle DAC$ [1 Mark]
In $\triangle ABC$ and $\triangle DAC$:

(1) $\angle BAC = \angle ADC$ (Given)

(2) $\angle ACB = \angle DCA$ (Common angle $\angle C$)

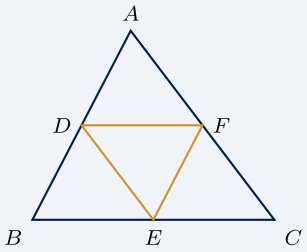
Therefore, by **AA similarity criterion**, $\triangle ABC \sim \triangle DAC$.

Step 2: Equating Corresponding Side Ratios [1 Mark]
Since corresponding sides of similar triangles are proportional:

$$\frac{CA}{CD} = \frac{CB}{CA} \implies \boxed{CA^2 = CB \cdot CD \quad \text{[Hence Proved]}}$$

Q9. In $\triangle ABC$, D, E, F are midpoints of sides AB, BC, CA respectively. Prove that $\triangle ABC$ is divided into four congruent triangles, each of which is similar to $\triangle ABC$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Given: In $\triangle ABC$, D, E, F are midpoints of AB, BC, CA respectively.
We use the Midpoint Theorem (a consequence of Converse of BPT) and SSS/AA rules:

Formula / Theorem Used: Line joining midpoints of two sides is parallel to and half of the third side.

Step 1: Applying Midpoint Theorem to All Pairs [1 Mark]
Since D, E, F are midpoints of AB, BC, CA:

- $DF \parallel BC$ and $DF = \frac{1}{2}BC = BE = EC$.
- $DE \parallel AC$ and $DE = \frac{1}{2}AC = AF = FC$.
- $EF \parallel AB$ and $EF = \frac{1}{2}AB = AD = DB$.

Step 2: Proving the Four Triangles Congruent [1 Mark]
Comparing each corner triangle with central triangle $\triangle EFD$:

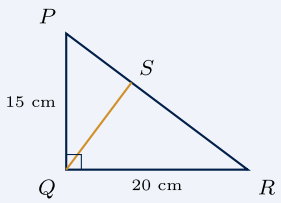
- In $\triangle ADF$ & $\triangle EFD$: $AD = EF$, $AF = ED$, $DF = FD \implies \triangle ADF \cong \triangle EFD$ (SSS).
- In $\triangle DBE$ & $\triangle EFD$: $DB = EF$, $BE = FD$, $DE = DE \implies \triangle DBE \cong \triangle EFD$ (SSS).
- In $\triangle FEC$ & $\triangle EFD$: $FE = EF$, $EC = FD$, $FC = DE \implies \triangle FEC \cong \triangle EFD$ (SSS).

Thus all four triangles are congruent: $\triangle ADF \cong \triangle DBE \cong \triangle FEC \cong \triangle EFD$.

Step 3: Similarity to $\triangle ABC$ [1 Mark]
In $\triangle ADF$ and $\triangle ABC$: $DF \parallel BC \implies \angle ADF = \angle B$ and $\angle AFD = \angle C$ (corresponding angles), so $\triangle ADF \sim \triangle ABC$ by AA similarity.
Since $\triangle DBE, \triangle FEC, \triangle EFD$ are all congruent to $\triangle ADF$ (hence equiangular with it), each of the four congruent sub-triangles is similar to $\triangle ABC$. **[Hence Proved]**

Q10. In the given figure, PQR is a right-angled triangle at Q. S is a point on PR such that $QS \perp PR$. If $PQ = 15$ cm and $QR = 20$ cm, find the length of QS. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: $\triangle PQR$ with $\angle Q = 90°$, $QS \perp PR$, $PQ = 15$ cm, $QR = 20$ cm.
We apply Pythagoras Theorem and AA Similarity (or Area Method):

Formula / Theorem Used: In right $\triangle PQR$, $PR^2 = PQ^2 + QR^2$ and $\triangle PQR \sim \triangle PSQ \implies \dfrac{QS}{QR} = \dfrac{PQ}{PR}$.

Step 1: Finding Hypotenuse PR [1 Mark]
By Pythagoras Theorem in right $\triangle PQR$:

$$PR^2 = PQ^2 + QR^2 = 15^2 + 20^2 = 225 + 400 = 625$$

$$\implies PR = \sqrt{625} = 25 \text{ cm}$$

Step 2: Proving $\triangle PQR \sim \triangle PSQ$ [1 Mark]
In $\triangle PQR$ and $\triangle PSQ$: $\angle PQR = \angle PSQ = 90°$ and $\angle QPR = \angle SPQ$ (common $\angle P$). Hence by AA similarity, $\triangle PQR \sim \triangle PSQ$.

Step 3: Calculating Altitude QS [1 Mark]

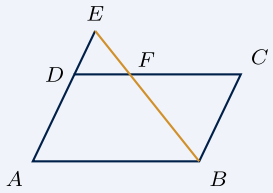
$$\frac{QS}{QR} = \frac{PQ}{PR} \implies \frac{QS}{20} = \frac{15}{25} = \frac{3}{5} \implies QS = \frac{3}{5} \times 20 = \boxed{\mathbf{12} \text{ cm}}$$

(Aliter: Area $= \frac{1}{2} PQ \cdot QR = \frac{1}{2} PR \cdot QS \implies 15 \times 20 = 25 \times QS \implies QS = 12$ cm.)

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Q11. In a parallelogram $ABCD$, E is a point on side AD produced and BE intersects CD at F. Prove that $\triangle ABE \sim \triangle CFB$. If $AB = 12$ cm, $CF = 8$ cm, and $FB = 6$ cm, find the length of line segment BE. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Given: Parallelogram $ABCD$; E lies on AD produced; BE meets CD at F; $AB = 12$ cm, $CF = 8$ cm, $FB = 6$ cm.
We use Parallelogram Angle Properties and AA Similarity:

Formula / Theorem Used: Opposite angles of a parallelogram are equal, and $AE \parallel BC \implies$ alternate interior angles with transversal BE are equal.

Step 1: Proving $\triangle ABE \sim \triangle CFB$ [2.5 Marks]
In $\triangle ABE$ and $\triangle CFB$:

(1) $\angle BAE = \angle FCB$ (Opposite angles of parallelogram $ABCD$: $\angle A = \angle C$)

(2) $\angle AEB = \angle CBF$ (Alternate interior angles, since $AE \parallel BC$ with transversal BE)

By **AA similarity criterion**, $\boxed{\triangle ABE \sim \triangle CFB}$ **[Hence Proved]**

Step 2: Calculating Length BE [2.5 Marks]
Since $\triangle ABE \sim \triangle CFB$, corresponding sides are proportional:

$$\frac{AB}{CF} = \frac{BE}{FB}$$

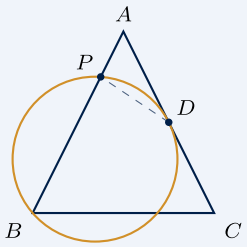
Substituting $AB = 12$ cm, $CF = 8$ cm, and $FB = 6$ cm:

$$\frac{12}{8} = \frac{BE}{6} \implies \frac{3}{2} = \frac{BE}{6} \implies BE = \frac{3 \times 6}{2} = \boxed{\mathbf{9}\text{ cm}}$$

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Q12. In $\triangle ABC$, $AB = AC$. A circle through B touches side AC at its midpoint D and intersects AB at P. Prove that: (i) $AD^2 = AP \times AB$, (ii) $AB = 4AP$. **[5]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]



Given: $\triangle ABC$ with $AB = AC$; circle through B is tangent to AC at its midpoint D ($AD = DC = \frac{1}{2}AC$) and cuts AB at P.
We apply Tangent-Chord Property (Alternate Segment Theorem) & AA Similarity:

> **Formula / Theorem Used:** The angle between a tangent (AD) and a chord (DP) equals the angle ($\angle PBD$) subtended by the chord in the alternate segment.

Part (i): Proving $AD^2 = AP \times AB$ [3 Marks]

Since AC touches the circle at D and DP is a chord through D, by the **Alternate Segment Theorem**: $\angle ADP = \angle PBD = \angle ABD$.

(Alternatively, for circle centre O: $OD \perp AC \implies \angle ADP = 90° - \angle ODP = \frac{1}{2}\angle POD = \angle PBD$.)

Now in $\triangle ADP$ and $\triangle ABD$:

(1) $\angle DAP = \angle BAD$ (Common angle $\angle A$)

(2) $\angle ADP = \angle ABD$ (Proved above)

By **AA similarity criterion**, $\triangle ADP \sim \triangle ABD$.
Corresponding sides are proportional:

$$\frac{AD}{AB} = \frac{AP}{AD} \implies \boxed{AD^2 = AP \times AB}$$

Part (ii): Proving $AB = 4AP$ [2 Marks]

Since D is the midpoint of AC and $AB = AC$, we have $AD = \frac{1}{2}AC = \frac{1}{2}AB$.

Substituting $AD = \frac{1}{2}AB$ into result (i):

$$\left(\frac{1}{2}AB\right)^2 = AP \times AB \implies \frac{AB^2}{4} = AP \times AB$$

Dividing both sides by $AB \neq 0$:

$$\frac{AB}{4} = AP \implies \boxed{AB = 4AP \quad \text{[Hence Proved]}}$$

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Q13. Case Study 11: High-Voltage Transmission Cable Crossing Mechanics

Two power transmission towers T_1 and T_2 of heights 15 m and 30 m stand vertically on horizontal ground separated by a span of 40 m. Tension guide wires run from the top of each tower to the anchored base of the opposite tower. Safety engineers must ensure the crossing clearance point P above ground meets high-voltage clearance norms. [4]

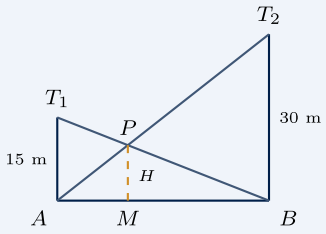
(i) **[1 Mark]** State the algebraic formula for crossing height H in terms of tower heights h_1 and h_2.

(ii) **[1 Mark]** Does the horizontal distance between the towers affect crossing height H? Explain.

(iii) **[2 Marks]** Calculate the exact vertical clearance of crossing point P above the level ground.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Geometric Setup: Let towers $AT_1 = h_1 = 15$ m and $BT_2 = h_2 = 30$ m stand vertically on level ground $AB = d = 40$ m. Guide wires T_1B and T_2A cross at P, and let $PM \perp AB$ with $PM = H$. Since AT_1, PM, BT_2 are all vertical, $AT_1 \parallel PM \parallel BT_2$.



We apply AA Similarity in the two overlapping right triangles:

Formula / Theorem Used: From $\triangle BMP \sim \triangle BAT_1$ and $\triangle AMP \sim \triangle ABT_2$, $\dfrac{H}{h_1} + \dfrac{H}{h_2} = \dfrac{BM + AM}{d} = 1.$

Sub-part (i): Algebraic Formula for Crossing Height H [1 Mark]

Since $\triangle BMP \sim \triangle BAT_1 \implies \dfrac{H}{h_1} = \dfrac{BM}{d}$ and $\triangle AMP \sim \triangle ABT_2 \implies \dfrac{H}{h_2} = \dfrac{AM}{d}$.

Adding gives $\dfrac{H}{h_1} + \dfrac{H}{h_2} = \dfrac{BM + AM}{d} = 1 \implies H\left(\dfrac{h_1 + h_2}{h_1 h_2}\right) = 1$, so:

$$\boxed{H = \frac{h_1 h_2}{h_1 + h_2}}$$

Sub-part (ii): Effect of Horizontal Span d on Crossing Height H [1 Mark]

No, the horizontal distance between the towers ($d = 40$ m) does **not** affect the crossing height H. Because $BM + AM = d$, the span d cancels identically ($\frac{BM+AM}{d} = 1$), so H depends solely on tower heights h_1 and h_2.

Sub-part (iii): Exact Vertical Clearance H Above Ground [2 Marks]

Substituting $h_1 = 15$ m and $h_2 = 30$ m into the formula:

$$H = \frac{15 \times 30}{15 + 30} = \frac{450}{45} = \boxed{\textbf{10 m}}$$

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